

Running a Debian Virtual Machine On Arch Linux

2025-11-01

1 Overview

component	name	version	notes
host kernel	Linux	6.17.6-arch1-1	
host OS	Arch Linux		
	QEMU	10.1.2	executable: /usr/bin/qemu-x86_64
	virsh	11.8.0	executable: /usr/bin/virsh
guest OS	Debian GNU/Linux	13 (trixie)	
	libguestfs	1.56.2	executable: /usr/bin/virt-copy-in

2 Setting Up a Virtual Machine (VM)

2.1 Get an Image For a VM

```
$ wget https://cloud.debian.org/images/cloud/trixie/20251006-2257/debian-13-nocloud-amd64-20251006-2257.qcow2
```

The download URL can be found from <https://cloud.debian.org/images/cloud> (accessed 2025-11-01).

```
$ file debian-13-nocloud-amd64-20251006-2257.qcow2
debian-13-nocloud-amd64-20251006-2257.qcow2: QEMU QCOW Image (v3), 3221225472 bytes (v3), 3221225472 bytes
```

2.2 Resize The Image From 3 GB To 32 GB

```
$ qemu-img create -f qcow2 -o preallocation=metadata debian-13-resized.qcow2 32G
```

```
$ virt-resize --expand /dev/sda1 debian-13-nocloud-amd64-20251006-2257.qcow2
debian-13-resized.qcow2
```

2.3 Create And Launch a VM

```
$ virt-install \
  --name debian-13 \
  --memory 4096 \
  --vcpus 2 \
  --disk ~/Downloads/debian-13-resized.qcow2 \
  --import \
  --os-variant debian13 \
  --network user
```

View available options for the `--os-variant` argument:

```
$ osinfo-query os | grep debian
```

In the guest, set Finnish keyboard layout by installing `console-setup`¹: The installer of version 1.240 includes an interactive keyboard layout selection.

¹<https://packages.debian.org/trixie/console-setup> (accessed 2025-11-01)

2.4 View, Shutdown And Restart a VM

```
$ virsh list --all
Id    Name           State
-----
1     debian-13      running
```

```
$ virsh shutdown debian-13
```

```
$ virsh start debian-13
$ virsh console debian-13
```

2.5 Take Snapshots Of a Shut Off VM, And Revert a VM To a Snapshot

```
$ virsh snapshot-create-as debian-13 000_init
$ virsh snapshot-list debian-13
$ virsh snapshot-revert debian-13 000_init
```

```
$ virsh snapshot-delete debian-13 000_init
```

2.6 Destroy a VM

```
$ virsh destroy debian-13
$ virsh undefine debian-13
```

3 Copying a File From Host To Guest

Install `qemu-guest-agent`² in the guest. An associated `systemd` service needs to be manually started once installed.

```
# systemctl start qemu-guest-agent
```

Shut down the guest. Then, on the host, use `virt-copy-in` from package `libguestfs` to copy a file to the guest.

```
$ virt-copy-in -d debian-13 ~/Downloads/hello-from-host-to-guest.txt /root/
```

²<https://packages.debian.org/trixie/qemu-guest-agent> (accessed 2025-11-01)